

Table S1. Meteorological data of the Gorongosa National Park in 2019: weather data from the dry season (May to September), the month of October and the three sampling periods in November: Dry (October 25th – November 5th), Intermediate (November 5th-15th), and wet sampling period (November 15th-25th). Historical weather data collected from the site Visual Crossing (2023). Average values of different weather factors, i.e. temperature data (Max, Min, Mean), relative humidity, precipitation probability, precipitation cover and cloud cover were compared among different periods (Dry season, October, and Dry, Intermediate and Wet sampling periods) with ANOVA - function ‘aov’, followed by pairwised Tukey tests in R, version 4.1.2. Shadowed cells indicate significant between the Dry season and each one of the other periods (light grey: $P < 0.01$; dark grey: $P < 0.001$); Letters “A” and “B” indicate the separation of significantly different groups within the sampling period: Dry vs. Intermediate vs. Wet ($P < 0.05$); bold numbers indicate the highly significant differences of Wet vs. Dry sampling periods ($p < 0.001$).

		Dry season		Sampling period (November)		
		May-September	October	Dry	Intermediate	Wet
Temperature	Max (°C)	28.7 (3.3)	33.6 (4)	37.9 (2.3)A	36.2 (2.2)A	32.1 (1.8)B
	Min (°C)	14 (2.5)	19.7 (2.2)	22.4 (1.5)A	23.1 (1.7)A	21.9 (1.1)A
	Mean (°C)	20.6 (2.8)	26.2 (2.8)	29.8 (1.6)A	29.4 (1.4)AB	26.6 (1.3)B
Relative humidity (%)		52.8 (12.5)	50.7 (12.5)	33.2 (9.1)A	45.5 (7.3)A	65.2 (1.1)B
Precipitation	Probability (%)	10.5 (30.7)	20.8 (41.5)	0	40 (51.6)A	86.7 (35.2)B
	Cover (%)	2.1 (8.2)	6.1 (16.6)	0	15.4 (25.2)A	46.4 (31.5)B
Cloud cover (%)		21.9 (19.9)	32.7 (28)	28.4 (23.6)A	58.8 (25.2)B	79.8 (10.6)B

Table S2. List of Caraboidea beetles (Cicindelidae, Carabidae) and their respective morphological traits regarding wing development (fully developed wings - macropterous, or apterous/brachypterous) and body size (cm). Data collected from literature specific of each species (Serrano et al., 2023) and/or confirmed/extracted from the actual specimens.

Species	Wings fully developed	Medium length (cm)
<i>Abacetus cursor</i>	Yes	5
<i>Abacetus denticollis</i>	Yes	9.25
<i>Abacetus discolor</i>	Yes	7
<i>Abacetus nigrinus</i>	Yes	10
<i>Abacetus percoides</i>	Yes	10.75
<i>Abacetus perturbator</i>	Yes	5.75
<i>Abacetus pseudomashunus</i>	Yes	14
<i>Anaulacus madagascariensis</i>	Yes	6.5
<i>Anthia alternata</i>	No	40.5
<i>Anthia burchelli petersi</i>	No	46.5
<i>Anthia circumscripta circumscripta</i>	No	44.5
<i>Anthia fornasinii fornasinii</i>	No	40.5
<i>Anthia omoplata</i>	No	38
<i>Apotomus annulaticornis</i>	Yes	3.5
<i>Apristus latipennis latipennis</i>	Yes	3.5
<i>Brachinus distans</i>	Yes	10
<i>Brachinus laetus</i>	Yes	8
<i>Brachinus leprieuri</i>	Yes	4
<i>Calosoma planicolle</i>	Yes	35
<i>Chlaenius conformis</i>	Yes	13
<i>Chlaenius cosciniophorus</i>	Yes	11.75
<i>Chlaenius cylindricollis</i>	Yes	14
<i>Chlaenius discopictus nuncius</i>	Yes	14.75
<i>Chlaenius dusaultii diagraphus</i>	Yes	16.5
<i>Chlaenius fulvicollis</i>	Yes	11.25
<i>Chlaenius kirki kirki</i>	Yes	15
<i>Chlaenius lugens</i>	Yes	18.75
<i>Chlaenius notabilis</i>	Yes	12
<i>Chlaenius paenulatus</i>	Yes	8
<i>Chlaenius peringueyi</i>	Yes	15.5
<i>Chlaenius zanzibaricus giganteus</i>	Yes	29
<i>Cimbionotum schueppelii</i>	Yes	4
<i>Crepidogaster langenhani</i>	No	10.75
<i>Crepidogaster longelineata</i>	No	4.65
<i>Crepidogaster protuberata</i>	No	9.75
<i>Cymindoidea regularis</i>	No	8.5
<i>Cypholoba alveolata ranzanii</i>	No	28
<i>Cypholoba graphipteroides bilunata</i>	No	25
<i>Cypholoba rutata</i>	No	22.5

<i>Cypholoba semisuturata vassei</i>	No	22.5
<i>Distichus bisquadrupunctatus</i>	Yes	11.5
<i>Distichus picicornis</i>	Yes	9.5
<i>Dromica dolosa latepolita</i>	No	9.9
<i>Eccoptoptera mutilloides mutilloides</i>	No	15.5
<i>Elaphropus aethiopicus</i>	Yes	2
<i>Elaphropus haemorrhoidalis</i>	Yes	2.5
<i>Elliptica compressicornis compressicornis</i>	Yes	17
<i>Galerita angustipennis</i>	No	22.5
<i>Graphipterus horni staudingeri</i>	No	12
<i>Graphipterus lineelus</i>	No	11.5
<i>Graphipterus tristis</i>	No	12
<i>Manticora latipennis</i>	No	52.5
<i>Megacephala asperata</i>	No	24.5
<i>Melaenus elegans</i>	Yes	9
<i>Melanchiton lucidulus</i>	Yes	10.5
<i>Mesolestes sp.</i>	Yes	2
<i>Mesolestes machadoi</i>	Yes	2.65
<i>Mesolestes nigrocephalus</i>	Yes	2.65
<i>Metagonum insolitum</i>	No	10.75
<i>Microlestes flavipes micromys</i>	Yes	2.65
<i>Microlestes zambezianus</i>	Yes	2.75
<i>Omostropus mandibularis</i>	Yes	8.5
<i>Parophonus tomentosus</i>	Yes	15
<i>Paussus cultratus</i>	Yes	5
<i>Paussus pseudoklugi</i>	Yes	5
<i>Pentaplatarthrus gestroi</i>	Yes	8
<i>Perigona schmitzi</i>	Yes	2.85
<i>Pheropsophus dregei</i>	Yes	17
<i>Pheropsophus insignis insignis</i>	No	15.5
<i>Pheropsophus mashunus</i>	No	14.5
<i>Pheropsophus stenopterus</i>	Yes	19
<i>Planetes quadricollis</i>	Yes	11
<i>Platytarus tessellatus</i>	No	6.5
<i>Prothymidia angusticollis</i>	No	9
<i>Scarites polyphemus</i>	No	26
<i>Scarites tenebricosus molossus</i>	No	30
<i>Siagona caffra</i>	Yes	22
<i>Siagona levasseuri</i>	Yes	15
<i>Siagona partita</i>	Yes	9
<i>Singilis africaorientalis kenyacus</i>	Yes	4
<i>Siopelus lucens</i>	Yes	9
<i>Styphlomerus neavei neavei</i>	Yes	10.5
<i>Systolocranius goryi</i>	Yes	20
<i>Tachys iridipennis</i>	Yes	2.7
<i>Tefflus carinatus carinatus</i>	No	35
<i>Tetragonoderus immaculatus</i>	Yes	5
<i>Triaenogenius carinulatus carinulatus</i>	No	20

Table S3. Summary of GLM analysis with the fixed effects of wing development (species with not fully developed wings – apterous/brachypterous - used as reference) on Caraboidea body size (response variable).

Response	Fixed effects	Estimate	Std. Error	t value	P value	F _(1,188)	P value
Body size	Macropterous spp.	-8.43	0.97	-8.67	< 0.001	105.9	< 0.001

Table S4. Tiger- and ground-beetle community found in grassland habitats in the intermediate and wet sampling periods, that were either exclusive species of grassland habitats in the dry period (light grey), or exclusive forest species in the dry period that appeared in grasslands in the intermediate-wet period (dark grey).

Species	Grassland habitats		Forested habitats	
	Dry period	Interm-Wet	Dry period	Interm-Wet
<i>Brachinus distans</i>	1	4	0	0
<i>Brachinus laetus</i>	1	1	0	0
<i>Brachinus lepriuri</i>	0	1	0	0
<i>Scarites polyphemus</i>	1	10	0	0
<i>Cimbionotum schueppelii</i>	0	1	0	0
<i>Apotomus sp.2</i>	0	1	0	0
<i>Tachys (Paratachys) sp.1</i>	0	2	0	0
<i>Abacetus cursor</i>	0	2	0	0
<i>Melanchiton lucidulus</i>	0	1	0	0
<i>Chlaenius paenulatus</i>	0	1	0	0
<i>Distichus bisquadrupunctatus</i>	0	1	1	1
<i>Elaphropus aethiopicus</i>	0	1	2	13
<i>Chlaenius conformis</i>	0	3	2	3
<i>Chlaenius peringueyi</i>	0	5	1	5
<i>Chlaenius zanzibaricus giganteus</i>	0	1	1	1
<i>Chlaenius notabilis</i>	0	7	2	15
<i>Chlaenius lugens</i>	0	4	1	1
<i>Graphipterus tristis</i>	0	1	19	4
<i>Anthia alternata</i>	0	3	12	8

References:

Serrano ARM, Baptista M, Carvalho R, Boieiro M, Mendes S, Bartz M, Timóteo S, Azevedo-Pereira HMVS, Aguiar CAS, Alves da Silva A, Alves J, Briones MJ, Sousa JP, Martins da Silva P (2023) Inventory of tiger- and ground-beetles (Coleoptera Caraboidea: Cicindelidae, Carabidae) in two sampling seasons of the Gorongosa National Park. Biodivers Data J (in press)

Visual Crossing (2023) Weather Data & API - Global Forecast & History Data.

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